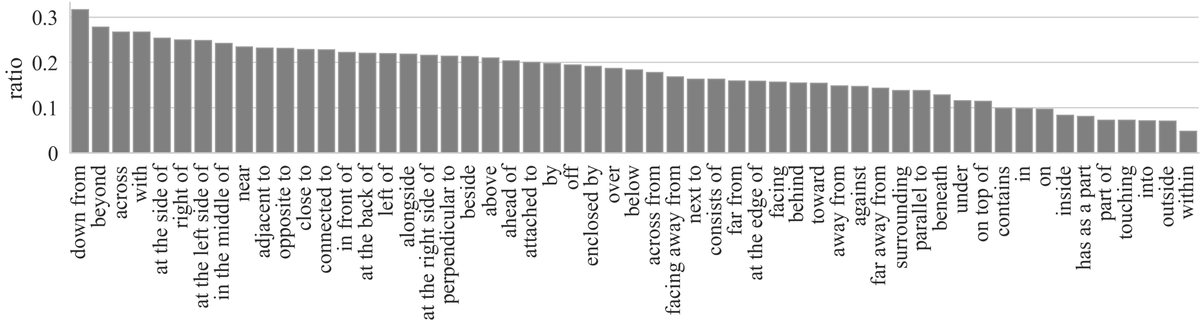


Figure 7: Per-relation probability of having two randomly chosen annotator disagreeing with each other (sorted from high to low). Only relations with >20 data points are included in the figure.

annotators. The relations that reflect the most basic relative coordinates of objects are also very frequent, e.g., "behind", "in front of", "on", "under", "at the left/right side of". Fig. 6 shows the distribution of concepts in the dataset. Note that the set of concepts is bounded by MS COCO and the distribution also largely follows MS COCO. Animals such as "cat", "dog", "person" are the most frequent. Indoor objects such as "dining table" and "bed" are also very dominant. In Fig. 6, we separate the concepts that appear at `ENT1` and `ENT2` positions of the sentence and their distributions are generally similar.

4.2 Where do annotators disagree?

While we propose using data points with high validation agreement for model evaluation and development, the unfiltered dataset is a valuable resource for understanding cognitive and linguistic phenomena. We sampled 100 examples where annotators disagree, and found that around 30 of them are caused by annotation errors but the rest are genuinely ambiguous and can be interpreted in different ways. This shows a level of intrinsic ambiguity of the task and variation among people.

Along with the validated VSR dataset, we also release the full unfiltered dataset, with annotators' and validators' metadata, as a second version to facilitate linguistic studies. For example, researchers could investigate questions such as where disagreement is more likely to happen and how people from different regions or cultural backgrounds might perceive spatial relations differently.

To illustrate this, the probability of two randomly chosen annotators disagreeing with each other is given for each relation in Fig. 7. Some of the relations with high disagreement can be interpreted in the intrinsic reference frame, which requires identifying the orientations of objects, e.g. "at the side of" and "in front of". Other relations have a high level of vagueness, e.g., for the notion of *closeness*: "near" and "close to". By contrast, part-whole relations, such as "has as a part", "part of", and in/out relations such as "within", "into", "outside" and "inside" have the least disagreement.

4.3 Case Study: Reference Frames

It is known that the relative reference frame is often preferred in English, at least in standard varieties. For example, Edmonds-Wathen (2012) compares Standard Australian English and Aboriginal English, as spoken by school children at a school on Croker Island, investigating the use of the relations "in front of" and "behind" when describing simple line drawings of a person and a tree. Speakers of Standard Australian English were found to prefer the relative frame, while speakers of Aboriginal English were found to prefer the intrinsic frame.

Our methodology allows us to investigate reference frame usage across a wide variety of spatial relations, using a wide selection of natural images. To understand the frequency of annotators using relative vs. intrinsic frames, we label instances' reference frames and study their distributions. The majority of examples that can be interpreted differently under different reference frames are left/right-related relations (i.e. "left/right of" and "at the left/right side of"). We find all left/right-related *true*[3] statements and classify them into three categories: (1) intrinsic (2) relative and (3) both (the caption is correct under either intrinsic and relative frames of reference). Among the 616 instances, 68 (11%) and 518 (84%) use intrinsic and relative frames respectively, 30 (5%) can be interpreted

[3]According to our guideline, false statements are interpreted as false under both frames.

with both frames. Since the vast majority of our annotators were native English speakers (91%), and all were university-educated, our finding is consistent with previous work suggesting that the relative frame is the most common frame in standard varieties of English.

Besides the overall trend, the use of reference frames can vary with the circumstances. Related patterns have been studied in cognitive science. For example, Vukovic and Williams (2015) find a three-way interaction between linguistic cues, spatial configurations in an image, and a person's own preferences on reference frames.

We investigated whether reference to a person in the image might influence how annotators comprehend the scene. 198 out of the 616 instances involve "person" in the caption. And out of the 198 human-involved instances, 32 (16%) use an intrinsic frame and 154 (78%) use a relative frame (12, i.e. 6%, can be interpreted with both frames), while the proportions were 9% and 87% for instances not involving "person". This is a statistically significant difference (using two-tailed Fisher's exact test, p=0.0054 if ignoring both-frame cases, and p=0.0045 if grouping both-frame and intrinsic cases). In other words, this suggests that the involvement of a human can more likely prompt the use of the intrinsic frame.

5 Experiments

In this section, we test VLMs on VSR. We first introduce baselines and experimental configurations in §5.1, then experimental results and analysis in §5.2. Then we discuss the role of frame of reference using experiments in §5.3 and finally conduct sample efficiency analysis in §5.4.

5.1 Baselines and Experiment Configurations

Baselines. For finetuning-based experiments, we test three popular VLMs: VisualBERT (Li et al., 2019)[4], LXMERT (Tan and Bansal, 2019)[5], ViLT (Kim et al., 2021)[6]. All three models are stacked Transformers (Vaswani et al., 2017) that take image-text pairs as input. The difference mainly lies in how or whether they encode the position information of objects. We report only finetuned results but not direct inferences from off-the-shelf checkpoints since some of their pretraining objectives are inconsistent with the binary classification task of VSR, thus requiring additional engineering.

We additionally test the alt text pretrained dual-encoder CLIP (Radford et al., 2021) as an off-the-shelf baseline model (no finetuning).[7] We follow Booth (2023) to construct negation or antonym of each individual relation. E.g., "facing" → "facing away from" and "ahead of" → "not ahead of". For each sample, we compare the embedding similarity of the image-caption pair and that of the negated caption. If the original pair has a higher probability then the model prediction is `True`, otherwise `False`. We call this method CLIP (w/ prompting). We only report direct prompting results without finetuning since CLIP finetuning is expensive.

Experimental configurations. We save checkpoints every 100 iterations and use the best-performing checkpoint on dev set for testing. All models are run three times using three random seeds. All models are trained with AdamW optimiser (Loshchilov and Hutter, 2019). The hyperparameters we used for training the three VLMs are listed in Table 3.

model	lr	batch size	epoch	token length
VisualBERT	2e-6	32	100	32
LXMERT	1e-5	32	100	32
ViLT	1e-5	12	30	max

Table 3: A listing of hyperparameters used for all VLMs ("lr": learning rate).

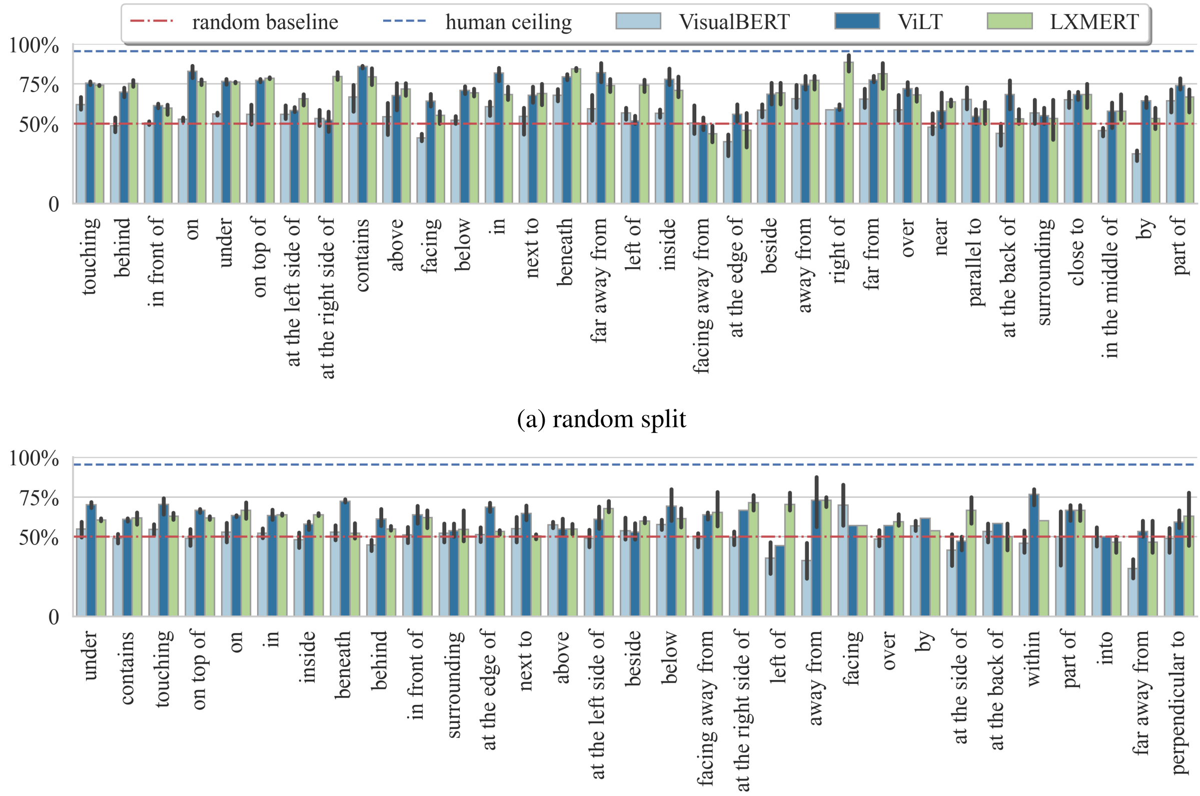
5.2 Experimental Results

model↓	random split	zero-shot split
human ceiling	95.4	
CLIP (w/ prompting)	56.0	54.5
VisualBERT	55.2$_{\pm 1.4}$	51.0$_{\pm 1.9}$
ViLT	**69.3**$_{\pm 0.9}$	**63.0**$_{\pm 0.9}$
LXMERT	**70.1**$_{\pm 0.9}$	61.2$_{\pm 0.4}$

Table 4: Model performance on VSR test set. CLIP is applied without finetuning but with carefully engineered prompts while the other three smaller models are finetuned on the training set.

In this section, we provide both quantitative and qualitative results of the four baselines. Through

[4] huggingface.co/uclanlp/visualbert-nlvr2-coco-pre
[5] huggingface.co/unc-nlp/lxmert-base-uncased
[6] huggingface.co/dandelin/vilt-b32-mlm
[7] huggingface.co/laion/CLIP-ViT-H-14-laion2B-s32B-b79K



(a) random split

(b) zero-shot split

Figure 8: Performance (accuracy) by relation on the random (upper) and zero-shot (lower) split test sets. Relation order sorted by frequency (high to low from left to right). Only relations with more than 15 and 5 occurrences on the random and zero-shot tests respectively are shown.

analysing the failure cases of the models, we also highlight the key abilities needed to solve this dataset.

As shown in Table 4, the best-performing models on the random split are LXMERT and ViLT, reaching around 70% accuracy while VisualBERT is just slightly better than the chance level. On the zero-shot split, all models' performance decline substantially and the best model ViLT only obtains 63.0% accuracy. The off-of-the-shelf CLIP model obtains around 55% on both sets, indicating its weaknesses in spatial reasoning echoing Subramanian et al. (2022)'s findings. Overall, these results lag behind the human ceiling by more than 25% and highlight that there is very substantial room for improving current VLMs.

Explicit positional information matters. Both LXMERT and ViLT outperform VisualBERT by large margins (>10%) on both splits. This is expected since LXMERT and ViLT encode explicit positional information while VisualBERT does not. LXMERT has position features as part of the input which encodes the relative coordinates of objects within the image. ViLT slices an image into patches (instead of object regions) and uses positional encodings to signal the patches' relative positions. VisualBERT, however, has no explicit position encoding. Bugliarello et al. (2021); Rösch and Libovický (2022) also highlight the importance of positional encodings of VLMs, which agrees with our observations.

Random split vs. zero-shot split. It is worth noting that the performance gap between the random and zero-shot splits is large. As we will show in §5.4, the underlying cause is not likely to be the number of training examples, but rather that concept zero-shot learning is fundamentally a challenging task. The gap suggests that disentangling representations of concepts and relations is challenging for current models.

Sensitiveness to random seeds. Model performance varies by about one to two percentage points. These fluctuations illustrate the importance of always reporting the average performance of multiple